What a buyer can learn from product, label and price data, and where to be careful

The same product listed by different sellers, with different titles and photos: matched automatically.

Wrong labels: caught for review, but not safe to auto-correct.

Price architecture and markdown timing: reconstructed from sales alone.

Matching is tested on 34,250 real listings. Labels and prices are tested on a simulated catalogue with a known answer key: that shows the method works, not what any retailer's market looks like.

Matching: text and image together find the same product best

On 34,250 real listings, combining the title and the photo puts the true match first for 74% of products and finds 79% of all true matches in the top 10.

Title alone puts it first for 63%, the photo alone for 62%: each catches matches the other misses.

A simulated catalogue and a small sample both favoured a simpler text method; the full set did not. Decisions should rest on full-scale tests.

Method	NDCG@10	Match first
Title + photo	0.766	74%
Title only (MiniLM)	0.639	63%
Photo only (CLIP)	0.602	62%
Title only (TF-IDF)	0.543	54%

Real listings: Shopee Price Match Guarantee (Kaggle, train split, non-commercial use). Zero-shot embeddings; nothing trained on Shopee.

Labels: colour and type are reliable, department is not

Department is the weak spot: many descriptions never say who the item is for, so men's and women's items get swapped.

Flagging disagreeing labels catches 95% of planted department errors but only 19% of flags are real errors. Use it as a review list.

Product type is easy in this simulation; do not read 99.8% as a real-world benchmark.

Attribute	Accuracy
Product Type	99.8%
Colour	91.6%
Department	75.7%

Simulated data (stylematch twin). Not real retailer data.

Price architecture: where each category sits and how often it is marked down

First prices are reconstructed from sales, not read from a price file.

In this simulation categories look almost the same by construction (similar prices and markdown rates), so there is no real category story here. The view is ready for real data.

Category	Median price	Marked down	Median depth
Coat	25	52%	40%
Dress	24	49%	30%
Shoes	24	46%	39%
Skirt	25	50%	31%
Sweater	25	50%	31%
Swimwear	24	47%	35%
T-shirt	26	53%	35%
Trousers	23	49%	35%

Simulated data (stylematch twin). Not real retailer data.

Markdowns: detectable from sales, and slow starters go first

Markdown detection: no false alarms and 89% of planted markdowns found; timing off by 0.4 weeks on average.

Sales in the first four weeks predict when a markdown arrives: error 3.6 weeks against 5.7 using the category average.

Caveat: this relationship was planted in the simulation. On real data it must be re-tested.

What this data cannot support

Forecasts: the simple recent-average beat seasonal models (error 12% vs 46%) because products launch and fade within the window. Two years is too short to learn seasons.

No competitor prices, stock levels or web-crawled listings, so sell-through is a proxy and not sell-out.

No causal price-elasticity claims: the simulation's price response is an assumption.

Next step: run on real catalogue data with real image embeddings and compare against these baselines.